

Workshop 3: Robust System Design and Project Management

Systems Analysis and Design

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1. Refined System Architecture

Our project focuses on predicting psychopathy levels based on Twitter behavior and linguistic patterns, using the dataset provided by Kaggle. After analyzing previous workshops, we refined our architecture to make it more modular, reliable, and ethically responsible.

The system is designed as a sequence of interconnected layers that transform chaotic, unstructured data into meaningful insights, while maintaining control and feedback to ensure stability.

Architecture Diagram

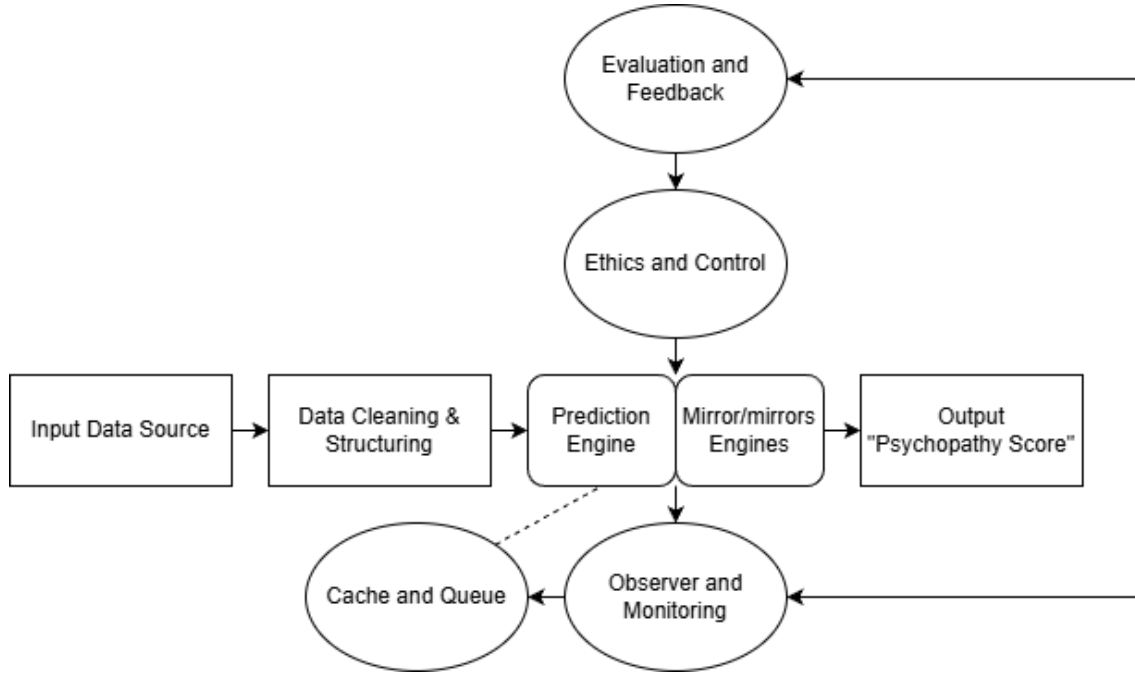


Figure 1: Refined Architecture for the Psychopathy Prediction System

How It Works

Our proposed system works as a structured flow that transforms noisy behavioral data into interpretable insights, while constantly monitoring its own stability, ethics, and accuracy. Below we explain how each layer works, why it exists, and how it interacts with others.

Input Data Source

This represents the system’s link to the real world — Twitter activity and language use. Since direct Twitter access is not available, this component is conceptual. In practice, we use the anonymized Kaggle dataset, which already contains 337 variables derived from linguistic and behavioral features.

Purpose: Simulate the chaotic external environment where information is born.

Reasoning: The data acts as the “environment” in a cybernetic sense — dynamic, incomplete, and full of uncertainty.

Data Cleaning & Structuring Layer

This layer cleans, organizes, and normalizes the dataset. It detects missing or extreme values, standardizes ranges, and ensures fairness across users. It serves as a filter of order, reducing noise and entropy.

Why it matters: Without this layer, the system would amplify noise, leading to chaotic or meaningless predictions.

Behavioral Feature Analysis

Once the data is structured, this module translates numbers and text into behavioral indicators. We divided it into three conceptual submodules:

- **Linguistic Analysis:** interprets emotional tone and word complexity.
- **Behavioral Profiling:** studies frequency and engagement metrics.
- **Social Structure Analysis:** observes relationships and follower patterns.

Why it matters: Isolated numbers become meaningful when analyzed together, revealing emergent behavioral patterns.

Prediction Engines

This is the core of the system — a set of regression models estimating psychopathy scores. Since only about 3% of the data represents high-psychopathy users, the model must deal with imbalance.

We included:

- A **main model** for primary predictions.
- One or more **mirror models** for redundancy and stability.

Why it matters: Reduces sensitivity to data fluctuations and keeps results stable.

Evaluation & Feedback Loop

This layer validates the model's performance using separate test data and sends corrective signals when necessary. If performance drops more than 5%, or overfitting appears, it recommends retraining or feature adjustment.

Why it matters: Maintains dynamic equilibrium — learning continuously without chaotic self-adjustments.

Ethics & Control Layer

This parallel layer guarantees ethical operation, ensuring privacy and fairness. It monitors all system interactions and blocks the use of variables that could cause bias or discrimination.

Why it matters: Keeps the system socially responsible and aligned with privacy standards.

Observer & Monitoring

This layer oversees the overall health of the system — latency, fairness, and drift. If it detects anomalies, it triggers maintenance or reports issues to the Ethics Layer.

Why it matters: Prevents unnoticed degradation or ethical drift over time.

Cache & Queue System

Handles operational rhythm and efficiency by storing temporary results and prioritizing tasks. Ensures logical order (e.g., evaluation after prediction) and prevents overloading.

2. Quality and Risk Analysis

Risk	Description	Mitigation
Data imbalance	Only 3% of data corresponds to high-psychopathy cases.	Apply resampling, weighting, and balanced evaluation metrics.
Feedback instability	Frequent retraining may cause oscillation.	Define stability thresholds and cooldown periods.
Ethical bias	Model may reinforce stereotypes.	Audit variables and use the Ethics Layer to filter sensitive data.
Latency or redundancy	Overlapping modules may slow data flow.	Add a shared Data Hub and prioritize queue execution.
Desynchronization	Reports might show outdated info.	Sync Observer updates with report refresh.

We will monitor performance weekly and keep historical comparisons to detect long-term drift or instability.

3. Project Management Plan

Team Roles

Role	Responsibility
David Sánchez Acero	Oversees documentation, system structure, and coordination across all components.
Juan David Bejarano Cristancho	Cleans and prepares datasets, manages feature analysis, and ensures data quality.
Raúl Andrés Díaz Losada	Designs, builds, and tunes regression models; ensures performance stability.
Juan David Ávila	Conducts evaluation, ethical reviews, and monitors quality control and reporting.

Methodology

We use a Kanban-style workflow to keep progress flexible and visible, using Trello for task management and Google Sheets for milestones.

Key Milestones

Phase	Deliverable	Date
System refinement	Updated architecture and diagram	Nov 6
Data preparation	Cleaned dataset	Nov 12
Modeling and validation	First regression results	Nov 18
Feedback & ethics testing	Stability and fairness report	Nov 22
Final integration	Prototype and report visuals	Nov 28

4. Incremental Improvements

From previous workshops we learned that a system must balance structure and adaptability. The initial design was static and lacked rhythm; now it integrates controlled feedback, ethical checks, and clearer data flow.

Main improvements include:

- Controlled feedback with thresholds to avoid instability.

- A common Data Hub to reduce redundancy.
- Defined roles and task priorities.
- Parallel Ethics & Monitoring layers.
- Clearer architecture diagram and modularity.

The system now behaves as a self-regulating structure rather than a static pipeline.

5. Reflection

Designing this system taught us that robustness is not about complexity, but about awareness. The architecture evolved from a simple sequence of processes into a dynamic organism that observes itself and reacts responsibly.

Although we cannot collect real-time Twitter data, we modeled how such a system could behave in theory — focusing on structure, ethics, and feedback. Ultimately, the system reflects how feedback-driven design helps maintain balance, even in complex, chaotic environments.
